

KIIT University – School of Computer Engineering

Capstone Project Report | Academic Year 2025–26

SAP SD — ORDER TO CASH (O2C)

Process Implementation

NovaMart Consumer Goods Pvt. Ltd.

SAP ECC 6.0 | FMCG / Retail | Order-to-Cash Module

Submitted By	Abhinav Raj – 2306369
Programme	B.Tech (Information Technology)
Industry	FMCG / Retail (Consumer Goods)
ERP System	SAP ECC 6.0 → S/4HANA (Phase 2)
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University	KIIT University, School of Computer Engineering

1. Problem Statement

NovaMart Consumer Goods Pvt. Ltd. — a mid-sized FMCG company with 3,200+ employees and 200+ distributors across 18 Indian states — operated on legacy Excel-based processes before SAP implementation. These manual systems led to critical operational failures across the Order-to-Cash cycle.

#	Problem	Impact	Root Cause
1	Manual Order Processing	Duplicate orders; 45 min/order processing time	Phone / email / WhatsApp orders
2	No Real-Time Inventory Visibility	Over-promising and under-delivery to distributors	Siloed warehouse & sales teams
3	Delayed Billing (5–7 days)	Poor cash flow and delayed AR collection	Manual invoice generation
4	No Centralized Customer Data	Inconsistent credit management	Isolated spreadsheets per team
5	Pricing Inconsistencies (12% error rate)	Revenue leakage and customer disputes	Manual discount decisions by reps
6	Poor Sales Analytics	No real-time dashboards for management	No centralized ERP system

2. Solution & Key Features

SAP Sales and Distribution (SD) module was implemented to automate the complete Order-to-Cash process — from customer inquiry through final payment. The solution integrates SAP SD with MM (Materials Management) and FI (Financial Accounting) for a seamless, real-time enterprise workflow.

2.1 O2C Process Flow — 8-Step Implementation

Step	Process	T-Code	Module	Key Output
1	Customer Inquiry	VA11	SAP SD	Inquiry Document
2	Quotation	VA21	SAP SD	Price Offer to Customer
3	Sales Order	VA01	SAP SD	Binding Order + Credit Check
4	Outbound Delivery	VL01N	SD / MM	Delivery + Picking List
5	Picking & Packing	VL02N	SD / WM	Warehouse Fulfillment
6	Post Goods Issue (PGI)	VL02N	SD / MM	Inventory Reduced, COGS Posted
7	Billing / Invoice	VF01	SAP SD	Auto FI Revenue Entry
8	Payment Receipt	F-28	SAP FI	AR Cleared, Cash Received

2.2 Core Features

- Automated Sales Order Processing (VA01) with ATP check, credit validation, and pricing determination.
- SAP Condition Technique for consistent pricing — conditions PR00, K007, HD00, MWST applied automatically.
- Real-time Credit Management (FD32) — automatic sales order block if credit limit is exceeded.
- SD ↔ MM Integration: Inventory deducted via Movement Type 601 at Goods Issue.
- SD ↔ FI Integration: Automatic revenue, AR, and GST postings on billing document creation.
- Centralized Customer Master (XD01) and Material Master (MM01) as a single source of truth.

3. SAP Screen Walkthrough (Process Screenshots)

This section illustrates key SAP transaction screens using structured descriptions mirroring the actual SAP GUI interface. Each screen represents a critical step in the NovaMart O2C cycle.

3.1 VA01 — Sales Order Creation

Screen Section	Field Values Entered
Order Type	OR (Standard Order)
Sales Area	NM01 / 10 (Wholesale) / 01 (Food & Beverages)
Sold-to Party	CUST-1001 — Star Retail Distributors Pvt. Ltd.
PO Number / Req. Delivery Date	SR/2025/0056 15-Jan-2025
Material / Qty	NM-FG-1001 50 CTN Delivering Plant: NM10
Pricing (auto-determined)	PR00: INR 480 K007: -24 HD00: +5 MWST: +82.98 = INR 543.98/CTN
System Output	'Standard Order 3000001 has been saved.' — ATP confirmed, credit check passed.

3.2 VL01N / VL02N — Outbound Delivery & Goods Issue

Screen Section	Actions & Values
Shipping Point	NM01-SP1 (Bhubaneswar Shipping Point)
Reference	Sales Order 3000001 — system auto-copies material & quantity.
Storage Location	SL10 — Finished Goods Store, Plant NM10
Picking (VL02N)	Picked Qty: 50 CTN Packed into 5 pallets (10 CTN each)
Post Goods Issue	Click 'Post Goods Issue'. System: 'Delivery 80000001: Goods issue posted.'
Auto Documents	Material Doc (inventory -50 CTN) + Accounting Doc (Dr. COGS 300000 Cr. FG Inventory 120000)

3.3 VF01 — Billing / Invoice Creation

Screen Section	Values & System Response
Billing Type	F2 — Standard Invoice
Reference	Delivery 80000001 — all billing items auto-populated.
Invoice Amount	50 CTN x INR 543.98 = INR 27,199.00 (incl. 18% GST)
FI Auto Posting	Dr. Customer AR (140000) INR 27,199 Cr. Revenue (800000) Cr. GST (220000)
Output	'Billing Document 90000001 saved.' Invoice PDF emailed to Star Retail.

3.4 F-28 — Incoming Payment Posting

Screen Section	Values Entered
Company Code	NM01 Currency: INR Payment Mode: RTGS
Bank Account	113000 (HDFC Current Account) Amount: INR 27,199
Open Item	Invoice 90000001 selected — Amount Assigned: INR 27,199
FI Posting	Dr. Bank (113000) Cr. Customer AR (140000) — AR Balance = 0

4. Technology Stack

The NovaMart O2C implementation leverages the full SAP ERP ecosystem. The core platform is SAP ECC 6.0 with a planned migration to SAP S/4HANA in Phase 2.

Layer	Technology	Role in O2C
ERP Core	SAP ECC 6.0 (ABAP Stack)	Central platform for all O2C transactions
Primary Module	SAP SD (Sales & Distribution)	Order management, delivery, billing
Integration – FI	SAP Financial Accounting (FI)	Revenue, AR, GST auto-postings
Integration – MM	SAP Materials Management (MM)	Inventory ATP check, goods issue (Mvt 601)
Integration – CO	SAP Controlling (CO / CO-PA)	Division-wise profitability analysis
Config Tool	SPRO (SAP IMG)	All configuration via Project Reference Object
User Interface	SAP GUI 7.70 + SAP Fiori (Phase 2)	Desktop GUI now; mobile Fiori planned
Data Migration	LSMW / BAPIs	Bulk upload of customer & material masters
Reporting	SAP Standard Reports + SAC (Phase 2)	VA05, VF05, FBL5N, S_ALR_87012172
Tax Engine	TAXINN (India Localization)	Automatic GST calculation via MWST condition
Output Management	SAP Smartforms / Adobe Output	Invoice PDF generation and email dispatch

5. Unique Points & Competitive Advantages

Feature	Description & Business Value
Three-Plant Organizational Design	NovaMart uses three SAP Plants (NM10 Bhubaneswar, NM20 Pune, NM30 Ahmedabad) enabling region-specific inventory management, independent storage locations, and realistic inter-plant stock transfer — unlike a simple single-plant setup.
Multi-Division Sales Hierarchy	Sales Organization NM01 spans three divisions (Food & Beverages, Personal Care, Home Care) allowing division-wise revenue analysis, independent pricing conditions per product line, and granular CO-PA profitability reporting.
Automated GST Compliance via TAXINN	India's TAXINN tax procedure integrates condition type MWST (18% GST) within pricing procedure RVAA01. Tax is calculated automatically on every SO and billing document, eliminating manual compliance risk and ensuring correct postings to GL 220000.
End-to-End Document Flow Integrity	Every O2C step creates a linked document chain: Inquiry → Quotation → SO → Delivery → GI → Invoice → Payment. SAP

	document flow (VA03) provides complete audit traceability, preventing orphan documents across 200+ distributors.
Credit Management with Automatic Block	FD32-based credit limits (INR 25L per customer) enforced at SO creation. Orders exceeding limits are auto-blocked. Pre-SAP: 8–10 credit breaches/month. Post-SAP: zero breaches recorded.
Real-World Business Simulation	Fully fictional-but-realistic company with actual Indian company codes, GST rates, INR currency, pan-India distribution network, and real FMCG product categories — making configuration directly applicable to real-world SAP SD projects.

6. Results & Performance Impact

Post-implementation metrics demonstrate measurable improvements across all key O2C performance indicators:

KPI	Before SAP	After SAP Implementation
Order processing time	45 minutes per order	8 minutes per order (82% reduction)
Billing cycle time	5–7 business days	Same day (post PGI)
Pricing error rate	12%	< 0.5% (condition technique)
AR visibility	Weekly manual Excel report	Real-time via FBL5N
Order fulfillment accuracy	82%	97%
Credit limit breaches	8–10 incidents/month	0 — auto-blocked by FD32
O2C cycle time (overall)	7–10 days	1–2 days (40% reduction target met)

7. Future Improvements

As Phase 2 of NovaMart's digital transformation, the following SAP enhancements are planned to extend O2C capabilities:

Initiative	Description & Business Value
SAP S/4HANA Migration	Migrate from ECC 6.0 to S/4HANA for real-time analytics via in-memory HANA database. Simplified data model (VBAK/VBAP → ACDOCA) enables sub-second O2C cycle reporting and embedded analytics.
SAP Fiori Mobile Applications	Deploy Fiori apps (Create Sales Order F0758, Track Deliveries, Billing Due List) so 200+ field sales reps can process orders from distributor locations without desktop SAP GUI access.
SAP CX / CRM Integration	Integrate SAP Sales Cloud (C4C) for lead management and opportunity tracking, automatically converting confirmed deals into SAP SD sales orders — closing the loop between marketing and order fulfillment.
SAP Transportation Management (TM)	Route optimization across NovaMart's 18-state network with carrier selection, freight cost calculation, and real-time shipment tracking — projected to reduce logistics costs by 15–20%.
SAP Analytics Cloud (SAC) Dashboards	Executive-level O2C dashboards with real-time KPIs: region-wise revenue, order backlog, AR aging, and forecast vs. actuals via Live Data Connection to S/4HANA.
AI/ML-Powered Demand Forecasting	SAP Integrated Business Planning (IBP) with ML demand signals to predict distributor order volumes by SKU and region, reducing stockouts and improving fulfillment accuracy beyond current 97%.

8. References

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Project at a Glance

Metric	Value
Company	NovaMart Consumer Goods Pvt. Ltd.
ERP Platform	SAP ECC 6.0 (Greenfield Implementation)
Module	SAP SD — Sales & Distribution
Integrations	SAP FI (Accounts Receivable), SAP MM (Inventory), SAP CO (Profitability)
Plants Covered	NM10 Bhubaneswar NM20 Pune NM30 Ahmedabad
Divisions	Food & Beverages Personal Care Home Care
Distributors	200+ across 18 Indian states
Key Achievement	O2C cycle reduced from 7–10 days to 1–2 days; 82% faster order processing